

Voltaic pile

Volta had discovered that two different metals submerged in acid could **generate a tiny electric current** between them. In 1800, he created his **voltaic pile**—the world's first battery. It was made from layers of copper, cardboard soaked in saltwater, and zinc—each group of three makes a “cell,” and **the more cells**, the greater the electric current produced. The current is generated by a **CHEMICAL REACTION** that moves electrons from one metal to the other.



Charging ahead

The voltaic pile was groundbreaking, but it leaked and didn't last long. The first **RECHARGEABLE** battery was the **lead-acid battery** invented in 1859 by French physicist Gaston Planté. The basic principle of this version remained unchanged into the 21st century. The first **dry battery** (using paste instead of liquid) was the zinc-carbon one created by German scientist **CARL GASSNER** in 1886, which led to the convenient batteries we put into flashlights today.

By the way...

It took me years of experimentation before I discovered the first practical method of generating electricity. The unit “volt” is named after me.

Zinc disk

Saltwater-soaked cardboard

Copper disk

How it changed the world

The battery was the first means of generating an electric current and the first easily portable energy source. Without it, there would be no phones, radio, or battery-operated devices.

Volta also discovered methane gas while poking around in a marsh, and invented an exploding pistol that he used to figure out the oxygen content of air.

The **ELECTRIC FLASHLIGHT** first came into use in 1899. Like today's flashlights, it used a dry-cell battery.



Tablet computers, **CELL PHONES**, and other mobile devices in everyday use all depend on a battery as a power source.

